

GarudaTwin

AI-Enabled Real-Time Digital Twin System for MALE UAV Propulsion Intelligence

Baseline Engine: Rotax 915 / 916 iS Turbocharged Aero Piston (1414 cc, Dual FADEC)

React 18 + Vite	PyTorch Bi-LSTM	XGBoost	FastAPI	Socket.io 100Hz
-----------------	-----------------	---------	---------	-----------------

Smart India Hackathon 2026 -- Problem Statement SIH26054

Design an AI-enabled real-time digital twin for MALE UAV engines to predict faults, compute remaining useful life, and enable autonomous mission replanning.

1. Project Overview

GarudaTwin is a full-stack, real-time Prognostics and Health Management (PHM) platform for MALE (Medium Altitude Long Endurance) UAV aero-piston engines. It creates a complete Digital Twin of the Rotax 915/916 iS engine running in a UAV platform (Predator/Heron class), enabling real-time fault detection, RUL prediction, SHAP explainability, and autonomous RTB replanning.

- Real-time engine health monitoring at 100 Hz via simulated CAN 2.0B bus
- Physics

2. System Architecture

Three independently running services communicate over localhost:

Service	Technology	Port	Role
Frontend GCS	React 18 + Vite 5 + Three.js + Tailwind	5173	Tactical Ground Control Station
Telemetry Engine	Node.js 18 + Express + Socket.io	5002	CAN Bus emulator & AI proxy
AI Microservice	Python 3.11 + FastAPI + Uvicorn	8001	AI/ML inference server

Node.js runs a 100 Hz physics simulation loop (setInterval 10 ms) and broadcasts at 20 Hz (setInterval 50 ms) via Socket.io. React queries the Python AI service at 1 Hz per live telemetry frame and falls back to a built-in first-principles physics bridge if needed. The Vite dev server proxies /api -> :5002 and /socket.io -> :5002 (WS) and /ai -> :8001.

3. CAN 2.0B Binary Frame Format

CAN ID	Bytes	Signal	Scale	Unit
0x100	0-1	RPM	x1 uint16 BE	RPM
0x100	2-3	Throttle	/100 uint16 BE	%
0x100	4-5	Fuel Flow	/100 uint16 BE	L/hr
0x100	6-7	Lambda (lambda)	/1000 uint16 BE	ratio
0x200	0-7	EGT Cyl 1-4	/10 per uint16 BE	degC
0x210	0-7	CHT Cyl 1-4	/10 per uint16 BE	degC
0x300	0-1	MAP	/1000 uint16 BE	bar
0x300	2-3	Oil Pressure	/1000 uint16 BE	bar
0x300	4-5	Oil Temp	/100 ? 50	degC
0x300	6-7	Vibration	/1000 uint16 BE	g-RMS

Gaussian Noise Model -- Box-Muller Transform

$$z0 = \text{sqrt}(-2 * \ln(u1)) * \cos(2*\pi*u2) \quad [u1,u2 \sim \text{Uniform}(0,1)]$$

$$X_sensor = X_true + z0 * \text{sigma}$$

Sensor	Noise sigma
RPM	4.0 RPM
MAP	0.005 bar
Vibration	0.008 g-RMS
EGT	1.2-1.5 degC
CHT	0.3 degC
Oil Pressure	0.02 bar
Oil Temp	0.1 degC
Generator Voltage	0.05 V

4. Physics First-Principles Engine -- All Formulas

4.1 ISA Atmosphere -- Pressure & Temperature

$$P_{\text{ambient}} [\text{bar}] = 1.01325 * (1 - 0.0225577 * h_{\text{km}})^{5.25588}$$

$$T_{\text{ambient}} [\text{degC}] = 15.0 - 6.5 * h_{\text{km}} \quad (\text{ICAO lapse rate})$$

$$h_{\text{km}} = h_{\text{ft}} * 0.0003048$$

4.2 Turbocharger Compressor Pressure Ratio

$$\alpha_{\text{WG}} = \min(1.0, \max(0.2, (\text{throttle}/100) * 1.15)) \quad [\text{wastegate target}]$$

$$PR_{\text{C}} = 1.0 + (\alpha_{\text{WG}} * 1.35) * (N / 5800) \quad [\text{compressor PR}]$$

$$MAP [\text{bar}] = \min(1.85, \max(0.70, P_{\text{ambient}} * PR_{\text{C}})) \quad [\text{Rotax 915iS limit}]$$

4.3 EGT Thermodynamic Balance (Combustion)

$$EGT_{\text{nominal}} [\text{degC}] = 820 + (\text{throttle}/100)*45 + (N/5800)*20 + (h_{\text{ft}}/10000)*8$$

$$\phi (\text{FAR}) = 0.88 + 0.12 * (\text{throttle}/100) \quad [\text{fuel-air equivalence ratio}]$$

$\phi < 1 \rightarrow$ lean $\rightarrow$ higher EGT; $\phi > 1 \rightarrow$ rich $\rightarrow$ lower EGT.

4.4 CHT Cooling ODE -- Liquid Cooling Balance

$$CHT_{\text{nominal}} [\text{degC}] = 102 + (\text{throttle}/100)*12 + (EGT_{\text{nominal}} - 800)*0.05$$

$$dT_{\text{EGT}}/dt = 0.80 * (T_{\text{target}} - T_{\text{EGT}}) \quad [\text{fast gas thermal response}]$$

$$dT_{\text{CHT}}/dt = 0.05 * (T_{\text{target}} - T_{\text{CHT}}) \quad [\text{slow metal mass, } \sim 20\text{s TC}]$$

$$dT_{\text{oil}}/dt = 0.02 * (T_{\text{target}} - T_{\text{oil}}) \quad [\text{very slow fluid mass}]$$

4.5 Lubrication System -- Hydrodynamic Oil Pressure

$$T_{\text{oil_nominal}} [\text{degC}] = 94 + (\text{throttle}/100)*8 + (N/5800)*5$$

$$P_{\text{oil_nominal}} [\text{bar}] = \max(2.2, 4.2 - (T_{\text{oil}} - 90)*0.018 - (5800 - N)*0.0002)$$

Physical effects: (1) viscosity-temperature dependency; (2) RPM-driven pump speed.

4.6 Structural Vibration Harmonic Baseline

$$V_{\text{baseline}} [\text{g-RMS}] = 0.22 + 0.16 * (N/5800) + (\text{throttle}/100)*0.05$$

4.7 Physics Residuals (Golden Model vs Actual)

$$R_{\text{EGT}_i} = EGT_{i_actual} - EGT_{\text{nominal}} \quad i=1,2,3,4$$

$$R_{\text{CHT}_i} = CHT_{i_actual} - CHT_{\text{nominal}}$$

$$R_{\text{MAP}} = MAP_{\text{actual}} - MAP_{\text{nominal}}$$

$$R_{\text{oil}_P} = P_{\text{oil_actual}} - P_{\text{oil_nominal}}$$

$$R_{\text{vib}} = V_{\text{actual}} - V_{\text{baseline}}$$

$$R_{\text{max}} = \max(|R_{\text{EGT}}|, |R_{\text{CHT}}|*2.5, |R_{\text{oil}_P}|*40, |R_{\text{vib}}|*80)$$

4.8 Real-Time Health Index from Residuals

$$\text{CRITICAL: } H = \max(15, 65 - R_{\text{max}}*0.45) \quad \text{if } R_{\text{max}} > 80 \text{ or } V > 1.2\text{g or } EGT_3 > 950 \text{ or } P_{\text{oil}} < 1.8$$

$$\text{DEGRADED: } H = \max(55, 88 - R_{\text{max}}*0.35) \quad \text{if fault active but below critical}$$

$$\text{NOMINAL: } H = 98.0$$

5. AI/ML Models

5.1 PyTorch Deep Autoencoder -- Anomaly Detection

Architecture: 12 -> [Linear(32) -> BN -> LeakyReLU(0.1)] -> Linear(16) -> LeakyReLU -> Linear(4) [z-latent]

Decoder: z(4) -> Linear(16) -> LeakyReLU -> Linear(32) -> LeakyReLU -> Linear(12) [reconstruction]

z_i	= (x_i - mu_i) / sigma_i	[z-score normalisation]
MSE_recon	= (1/12) * sum_i (z_i - z_hat_i)^2	
s_anomaly	= min(1.0, MSE_recon / 0.35)	
is_anomaly	= MSE>0.085 OR max R_EGT >55 OR R_vib >0.45 OR P_oil<1.8	
SHAP_w_i	= feat_err_i^2 / sum_j(feat_err_j^2) * 100%	

5.2 PyTorch Bi-LSTM -- RUL Prediction

Input: (1, 15, 12) -> BiLSTM(hidden=48, layers=2, bidirec) -> 96-dim last hidden state

Branch 1 (Mean RUL): FC(96->32) -> ReLU -> FC(32->1) | Branch 2 (Variance): FC -> Softplus

seq_t	= z_norm * (0.96 + 0.04*(t/15))	t=0..14 [synthetic temporal slope]
d_factor	= 1.0 + EGT_excess*0.04 + V_excess*4.5 + P_oil_loss*3.2	
RUL [hr]	= max(0.4, 842.0 / d_factor)	
sigma_CI	= sqrt(max(0.05, Var_LSTM) + RUL*0.08)	
CI_95	= 1.96 * sigma_CI	[95% Confidence Interval]

5.3 IsolationForest (Production) -- Anomaly Score

s_anomaly	= 1 / (1 + exp(s_IF * 12))	[logistic sigmoid on IF decision fn]
is_anomaly	= TRUE if s_IF < -0.5	

5.4 XGBoost 7-class Fault Classifier

Class	Fault Mode
none	Nominal operation
injector	Cylinder injector restriction/clog
lubrication	Oil pump/lubrication fault
cooling	Cooling system degradation
misfire	Combustion misfire/ignition fault
vibration	Structural bearing fault
sensor_drift	Sensor bias/calibration drift

Training: Weighted F1 = 0.9368 | 71 rolling-window features | C-MAPSS + Physics-synthetic data

5.5 XGBoost RUL Regressor

RUL_ML [hr]	= XGBoost_prediction_seconds / 3600
-------------	-------------------------------------

Validation: MAE = 63.6 seconds | RMSE = 108.3 seconds | 71 features

6. Multi-Physics Fatigue Stress Decomposition & RUL

6.1 Thermal Stress $S_{thermal}$

$$S_{thermal} = 1.0 + \max(0, (CHT-118)/14)^{1.8} + \max(0, (T_{oil}-95)/12)^{1.6*0.7} + \max(0, (EGT-880)/25)^{1.5*0.6}$$

6.2 Mechanical Stress $S_{mechanical}$

$$S_{mechanical} = 1.0 + \max(0, (V - 0.35) / 0.22)^{2.0} \quad [V \text{ in g-RMS}]$$

6.3 Lubrication Stress $S_{lubrication}$

$$S_{lubrication} = 1.0 + \max(0, (2.8-P_{oil})/0.8)^{2.2} + 1.4 \text{ if } T_{oil}>110 \text{ AND } P_{oil}<2.5$$

6.4 Combustion Stress $S_{combustion}$

$$S_{combustion} = 1.0 + (EGT - 920) / 30 \quad [\text{only when } EGT > 920\text{degC}]$$

6.5 Operating Load Stress $S_{operating}$

$$S_{operating} = 1.0 + ((N-5200)/600)*0.5 * I(N>5200) + ((MAP-1.65)/0.25)*0.4 * I(MAP>1.65)$$

6.6 Combined Stress (Multiplicative Model)

$$S_{combined} = S_{therm} * S_{mech} * S_{lub} * S_{comb} * S_{op} * M_{fault} * F_{anomaly}$$

$$M_{fault} = 2.4 \text{ if non-nominal fault active (else } 1.0)$$

$$F_{anomaly} = 1.0 + \max(0, s_{anomaly}) * 3.2$$

6.7 Engine Degradation Index (EDI 0-100)

$$EDI = \max(0, \min(100, (S_{combined}-1)*14 + s_{anomaly}*22 + (M_{fault}-1)*8))$$

6.8 Per-Subsystem Degradation Scores

Subsystem	Formula
Thermal	$\min(100, (S_{thermal}-1)*48)$
Mechanical	$\min(100, (S_{mech}-1)*52)$
Lubrication	$\min(100, (S_{lub}-1)*38)$
Combustion	$\min(100, (S_{comb}-1)*42)$
Fuel	$\min(100, Q_{fuel}-26 /26*60 + s_{anomaly}*10)$
Electrical	$\min(100, (S_{elec}-1)*55)$

6.9 Health Index Score & RUL

$$H = \min(100, \max(12, 100 - s_{anomaly}*48 - (S_{combined}-1)*3.5 - (M_{fault}-1)*12))$$

$$D_{nominal} = 0.045 \text{ \%/hr} \quad (\text{nominal Rotax TB0 degradation rate})$$

$$D_{active} = D_{nominal} * S_{combined}$$

$$RUL_{phys} = (H - 50) / \max(0.04, D_{active}) \quad [\text{hours; } 50\% = \text{MEL threshold}]$$

$$RUL_{final} = \max(2.0, \min(RUL, (2000 - h_{total})*0.85)) \quad [2000h = TB0]$$

7. Kalman Filter -- RUL State Estimation

A 1D Kalman filter smooths the raw RUL estimate, with dynamic process noise linked to combined stress.

Predict Step		
x_pred	= x_hat_(k-1)	[state prediction = last estimate]
P_pred	= P_(k-1) + Q	[covariance prediction]
Update Step (Measurement = raw RUL)		
K_k	= P_pred / (P_pred + R)	[Kalman Gain]
x_hat	= x_pred + K * (z_k - x_pred)	[state update]
P_k	= (1 - K) * P_pred	[posterior covariance]
Parameters		
Parameter	Value	Description
x0 (initial state)	2000 hours	Full TBO life
P0 (initial variance)	100	High initial uncertainty
Q (process noise)	0.05 + (S_combined-1)*0.5	Dynamic -- stress raises uncertainty
R (measurement noise)	10	Model prediction uncertainty
RUL_lower	= x_hat - 1.96*sqrt(P_k)	[95% CI lower bound]
RUL_upper	= x_hat + 1.96*sqrt(P_k)	[95% CI upper bound]

8. Weibull Failure Risk Model

P_failure = 1 - exp(-(EDI / 55)^beta)	
beta = 2.0	if Rapidly Degrading [wear-out Weibull shape]
beta = 1.5	otherwise [gradual degradation regime]
P_failure Score	
Risk Level	
< 0.05	LOW
0.05 - 0.20	MEDIUM
0.20 - 0.50	HIGH
> 0.50	CRITICAL
Multi-Horizon Risk Projection	
H_proj(Dt)	= max(0, H - D_active * Dt)
EDI_proj(Dt)	= min(100, EDI + D_active * Dt * 18)
P_risk(Dt)	= 1 - exp(-(EDI_proj/55)^beta) [Dt = 1h, 4h, 8h, 24h]

9. Explainable AI -- SHAP Attributions

9.1 Autoencoder SHAP-Style Attribution (Pipeline A)

```
w_i    = feat_err_i^2 / sum_j(feat_err_j^2) * 100%
w_vib  = min(80, |R_vib|*45)  [injected separately if |R_vib| > 0.3]
```

9.2 Physics-Residual Attribution (Pipeline B)

```
dev_i   = |x_i - x_i_nominal| / x_i_nominal * 100%
import_i = dev_i / sum_j(dev_j) * 100%
```

Direction: 'increases_risk' if parameter deviates in the degradation direction; else 'normalizing'.

9.3 Physics Explanation Map

Feature	Physics Root-Cause Explanation
EGT_Cyl3	EGT +120degC residual -> lean-burn combustion: injector clog / vapor lock
Vibration_gRMS	Broadband >1.0g -> rotational asymmetry / combustion imbalance
Oil_Pressure	Film < 2.0 bar -> journal bearing lubrication integrity at risk
MAP	Pressure-vs-throttle mismatch -> wastegate actuator hysteresis / boost leak
RPM	Cyclical hunting -> outside FADEC governor closed-loop bandwidth

10. RL Mission Replanner

10.1 Haversine Great-Circle Distance

```
a    = sin^2(Deltaphi/2) + cos(phi1)*cos(phi2)*sin^2(Deltalambda/2)
d    = 2*R*atan2(sqrt(a), sqrt(1-a))  [R=3440.065 NM]
```

10.2 RL Policy Decision Table

Health Index	Throttle	RPM	V/S (fpm)	Destination
< 40%	58% MCP	4200	?350	Nearest emergency strip
40-70%	68% MCP	4600	?200	AFS Uttarlai (Home)
> 70%	78% MCP	4850	0	AFS Uttarlai (Home)

Emergency divert trigger: RUL < 1.2 hours OR Health < 35%

10.3 Waypoint Interpolation

```
phi_wp_i    = phi_curr + (i/N)*(phi_dest - phi_curr)
lambda_wp_i = lam_curr + (i/N)*(lam_dest - lam_curr)
h_wp_i      = h_curr   - (i/N)*(h_curr   - h_dest)  [ft]
t_flight    = (d_NM / V_cmd) * 60  [minutes]
SafetyMargin = RUL_hours / (t_flight/60)
```

11. Fault Injection System -- 7 Fault Modes

CYL3_INJECTOR -- Cylinder 3 Injector Restriction
EGT3 += 135*sev + sin(t*8)*12*sev -> >970degC CHT3 += 28*sev -> >135degC Vibration += 0.95*sev -> >1.2g Lambda += 0.18*sev (lean shift)
BLOW_BY -- Piston Ring Blow-By
T_oil += 32*sev -> >130degC P_oil -= 1.65*sev -> ~2.2 bar Vibration += 0.65*sev
OIL_PUMP_CAVITATION -- Oil Aeration / Relief Valve Flutter
P_oil -= 2.3*sev + sin(t*15)*0.75*sev -> <1.5 bar Vibration += 1.35*sev + N(0,0.2*sev) -> bearing distress >1.6g
TURBO_WASTEGATE_STUCK -- Wastegate Jammed
MAP += 0.58*sev -> ~2.0 bar (overboost) All EGT += 42-48*sev RPM += 350*sev
COOLING_DEGRADATION -- Radiator Fouling
All CHT += 32-36*sev -> all cylinders overheat T_coolant += 35*sev -> coolant boilover
GENERATOR_FAILURE -- Alternator Field Fault
V_bus -= 4.9*sev -> ~23.5V (battery only) I_gen -= 33.2*sev -> load shedding
PRGB_DEGRADATION -- Propeller Gearbox Gear Wear
Vibration += 2.15*sev + N(0,0.15) -> severe broadband vibration

12. Feature Engineering Pipeline -- 71 Features

```
F_total = n_sensors + n_sensors * n_windows * n_stats
          = 14 + 14*2*2 = 14 + 56 = 71 (including health_index)

rolling_mean_w(t) = (1/w) * sum_{i=t-w+1}^{t} x_i
rolling_std_w(t)  = sqrt((1/w) * sum_i (x_i - mean_w)^2)
```

Windows: 30 timesteps (short-term trend) and 60 timesteps (medium-term trend) at 1 Hz sampling.

14 raw sensors: rpm, true_cht, sensor_cht, egt, oil_pressure, oil_temp, fuel_flow, vibration, battery_voltage, injection_timing, health_index, altitude, ambient_temp, throttle. Each gets: raw + rmean30 + rstdd30 + rmean60 + rstdd60 = 5 features each -> 70 + health_index = 71.

13. Validation Metrics & System Performance

Model	Metric	Value
XGBoost Fault Classifier	Weighted F1	0.9368
XGBoost RUL Regressor	MAE	63.6 seconds
XGBoost RUL Regressor	RMSE	108.3 seconds
IsolationForest	Precision	0.7344
IsolationForest	Recall	0.0607
IsolationForest	ROC-AUC	0.6992
PyTorch Autoencoder	Anomaly Threshold (MSE)	0.085
Unified Pipeline	Inference Latency	~256-290 ms

Component	Rate / Latency
Physics simulation loop	100 Hz (10 ms setInterval)
Telemetry broadcast to browser	20 Hz (50 ms setInterval)
AI/ML inference cadence	1 Hz (per live telemetry frame)
RL mission replan	< 200 ms
3D WebGL rendering	60 FPS
Unified pipeline latency	~256-290 ms

14. API Reference

Python FastAPI (Port 8001)		
Method	Endpoint	Description
GET	/health	Service status + model inventory
POST	/detect-anomaly	PyTorch Autoencoder MSE anomaly
POST	/predict-rul	Bi-LSTM RUL + 95% CI
POST	/explain-shap	SHAP attribution + physics explanation
POST	/rl-replan	RL RTB trajectory optimisation
POST	/api/health-rul/predict	UNIFIED: IsolationForest + XGBoost + Kalman + Weibull
POST	/api/health-rul/detect-anomaly	IsolationForest anomaly only
POST	/api/health-rul/predict-rul	XGBoost RUL only
POST	/api/health-rul/explain-shap	TreeSHAP attributions
POST	/api/health-rul/advisory	Maintenance advisory output

Node.js Express (Port 5002)		
Method	Endpoint / Event	Description
GET	/api/health	Telemetry engine status
POST	/api/faults/inject	Inject fault (type + severity 0-1)
POST	/api/faults/clear	Clear all faults -> nominal
POST	/api/health-rul/predict	Proxy -> Python :8001
WS	telemetry_frame	20 Hz real-time telemetry push
WS	inject_fault	Socket: inject fault
WS	clear_fault	Socket: clear fault
WS	update_manual_conditions	Manual mission param override

15. Fleet Management -- Vahak UAV Squadron

UAV	Role	Health	RUL (hrs)	Flt Hrs	Status
Vahak-1	Active Testbed (Live Twin)	LIVE	LIVE	1248.6	LOITER
Vahak-2	Escort Lead	96.2%	785	415.0	ON STATION
Vahak-3	Relay Orbit	99.1%	1120	80.0	CLIMB TO CRUISE
Vahak-4	Perimeter Patrol	84.5%	420	780.0	DERATED CRUISE
Vahak-5	Hangar Reserve	72.0%	120	1080.0	GND MAINT

Maintenance Advisory State Machine		
EDI Range	Action	Urgency
0 - 10	MONITOR	Continue Monitoring
10 - 28	INSPECTION	Inspection Advised (5-10h)
28 - 48	MAINTENANCE	Maintenance Required (24h)
48 - 68	MISSION_RESTRICTION	Ground before next flight
> 68	ENGINEERING_REVIEW	Remove from service immediately

Disclaimer & References

This system is an AI-assisted research and competition prototype. All prognostics outputs are for R&D advisory purposes only and are NOT certified aircraft maintenance data. System follows the Rotax 915/916 iS Aircraft Maintenance Manual (AMM) reference guidelines. It does NOT replace certified PHM systems, licensed avionics engineers, or Type Certificate data.

References

- Rotax 915 iS / 916 iS Operator's Manual -- BRP-Rotax GmbH

• C-MAP